Jian-Kai Zhu

Graphics / Rendering Engineer

Taipei, Taiwan | Email | Github | Portfolio | +886 963-674-796

Summary

Graphics / Rendering Engineer with experience in real-time rendering, Direct3D 11/12, GPU resource management, and graphics debugging. Experienced in building graphics API interception and GPU data extraction systems for commercial games, with a strong background in computer graphics and 3D research.

Technical Skills

Programming:

- C++, Python, Rust

Graphics/Rendering:

- Direct3D 11/12, HLSL, Vulkan, OpenGL/GLSL
- Real-time Rendering, PBR, Ray Tracing

Graphics Tools:

- RenderDoc, ReShade, Blender

GPU/Systems:

- GPU Resource Management
- GPU-CPU Synchronization
- GPU Readback
- Performance Profiling

Professional Experiences

Shanda Corp. **Roppongi, Japan**
Independent Contractor — Graphics/World Model Data Infrastructure *Feb 2026 - Present*

- Develop a real-time **D3D11/D3D12 graphics API interception system** for extracting general-purpose extraction of in-game camera pose, metric depth, and object-level scene information from commercial games without source-code access.
- Analyze rendering pipelines using **RenderDoc**, **shader inspection**, **resource bindings**, and **draw/indirect draw calls** to identify, locate, and validate extractable game data.
- Implement **GPU-to-CPU resource readback and frame synchronization** using staging/readback resources, ring buffers, and GPU fences to maintain frame-aligned data capture under variable GPU workloads.
- Reconstruct metric depth from normalized GPU depth buffers using recovered camera projection parameters, and build synchronized visualization tools for camera, depth, and object information.
- Adapt the extraction pipeline across **8 commercial games** spanning Unreal Engine, Anvil, Snowdrop, RAGE, and proprietary engines.

Institute of Information Science, Academia Sinica **Taipei, Taiwan**
Research Assistant | Supervisor: Dr. Jen-Chun Lin *Sep 2023 - Present*

- Developed a Blender add-on for animation retargeting and visualization of text-to-motion synthesis results.
- Developed a text-to-motion generation framework improving semantic alignment and motion fidelity, published in Computers & Graphics.

One Stop Studio **Taipei, Taiwan**
Intern Software Engineer *Sep 2021 - June 2023*

- Developed a commercial BIM application built on Blender 2.79, working on tools for architectural component placement, aggregation, and pricing.
- Optimized rendering performance for interactive visualization and editing of large-scale scenes with many components.
- Investigated rendering issues using frame capture and graphics debugging tools.

Publications

Tai-Chi: Text-to-Motion Generation with Locality-Aware Bipartite Body-Part Motion Prior
Jian-Kai Zhu, Wen-Li Wei, Jen-Chun Lin
in press of **Computers & Graphics 2026.**

StyleFaceUV: a 3D Face UV Map Generator for View-Consistent Face Image Synthesis
Wei-Chieh Chung*, Jian-Kai Zhu*, I-Chao Shen, Yu-Ting Wu, Yung-Yu Chuang (*: joint first authors)
in proceeding of **British Machine Vision Conference (BMVC) 2022**

Cycle-Consistent Motion Refinement for Speed-Controllable Text-to-Motion Generation
Jian-Kai Zhu, Wen-Li Wei, Jen-Chun Lin
submitted to **ICASSP 2027, under review**

Education

National Taiwan University **Taipei, Taiwan**
M.S. in Computer Science
2021 - 2023 | **GPA**: 4.18/4.3
Thesis: Full-body Image Inversion on GAN Latent Space with Improved Training Procedure

National Taiwan Normal University **Taipei, Taiwan**
B.Eng. in Computer Science
2017 - 2021 | **GPA**: 4.2/4.3, Ranking: 1/53 (Top 5%)